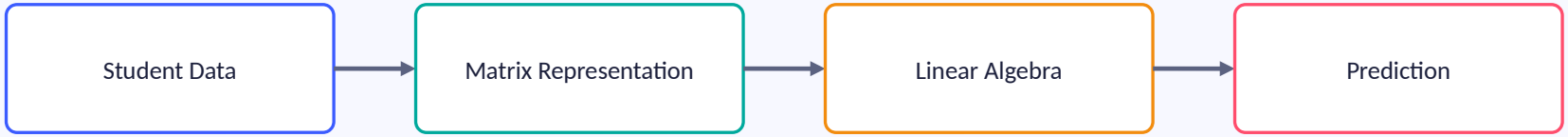
Student Performance Prediction Using Linear Algebra

Applying RREF, Gram-Schmidt, QR Decomposition, Projection and Least Squares

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Problem Statement

- Student performance depends on many academic and social factors
- Goal: predict the final grade G3
- Look at the mathematics behind the prediction, not a black box

studytime	failures	absences	G1	G2	freetime	goout	G3
2	0	6	5	6	3	4	6
2	0	4	5	5	3	3	6
2	3	10	7	8	3	2	10
3	0	2	15	14	2	2	15

Features

studytime · failures · absences · G1 · G2 · freetime · goout

Target: G3 — final grade

Dataset preview is illustrative — UCI Student Performance

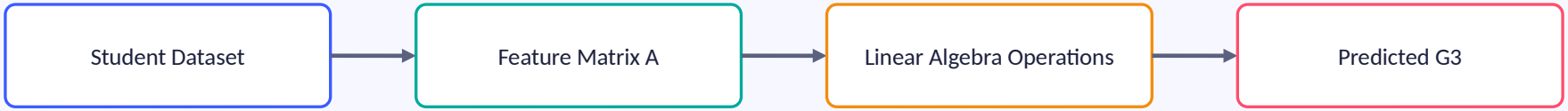
Converting the Dataset into a Matrix

$$Ax \approx b$$

- A — feature matrix
- x — unknown coefficients
- b — actual final grades
- Ax — predicted final grades

$$A = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$$

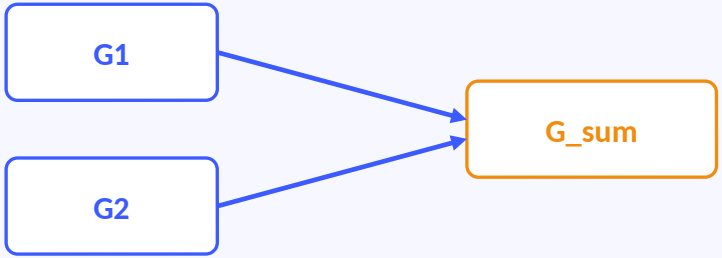
First column of 1s = intercept



Matrix form lets us apply linear algebra directly.

Independent vs Redundant Information

$$G_sum = G1 + G2 \quad \Rightarrow \quad G1 + G2 - G_sum = 0$$



G_sum is exactly dependent on G1 and G2.

$$\text{rank}(A) = 8$$

$$\text{columns} = 9$$

$$\text{nullity} = 1$$

- Rank — independent columns
- Nullity — dependent directions
- Pivot columns — independent
- Free columns — redundant

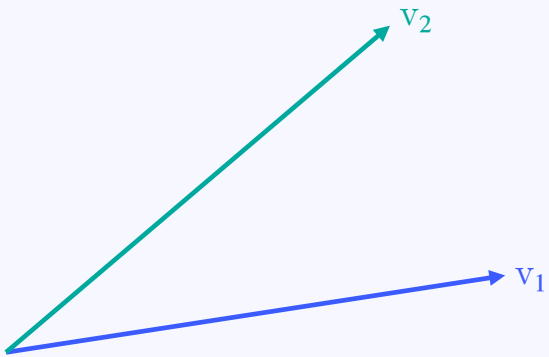
$$\text{rank}([A|b]) > \text{rank}(A)$$

No exact solution to Ax = b → use least squares

Creating an Independent Basis

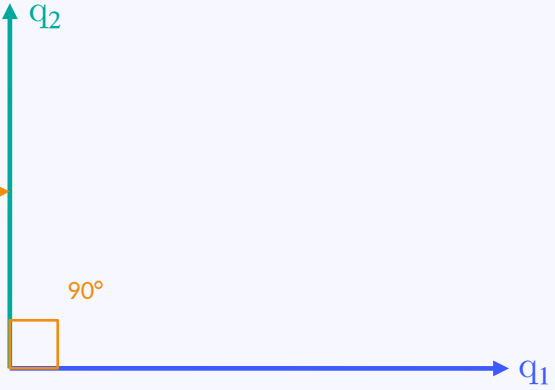
Remove redundant G_sum. G1 and G2 are still strongly correlated — Gram-Schmidt makes them perpendicular.

Original vectors



Gram-Schmidt

Orthogonal vectors



$$A = QR$$

Q: orthonormal basis · R: upper triangular

Finding the Best Approximation

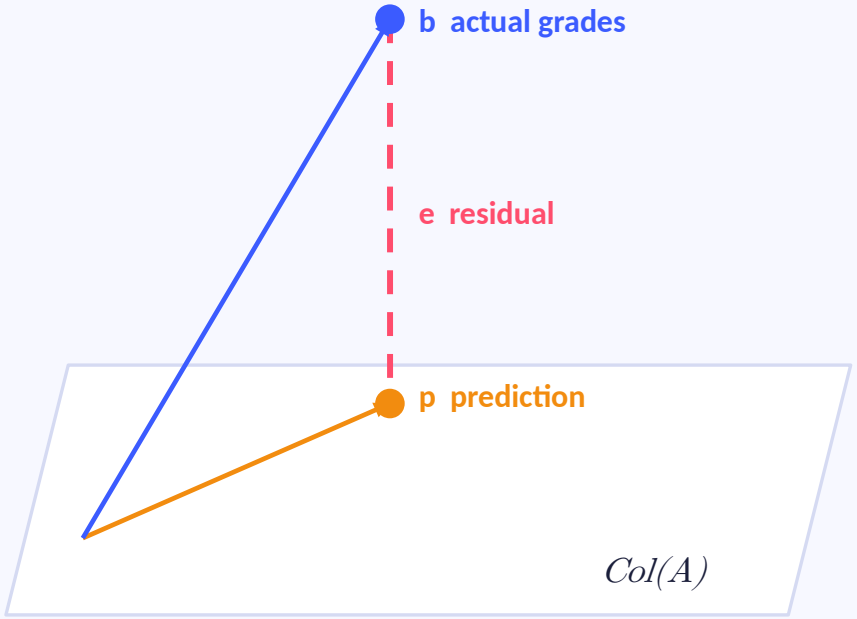
$$p = QQ^T b$$

$$e = b - p$$

$$A^T e = 0$$

$$Rx = Q^T b$$

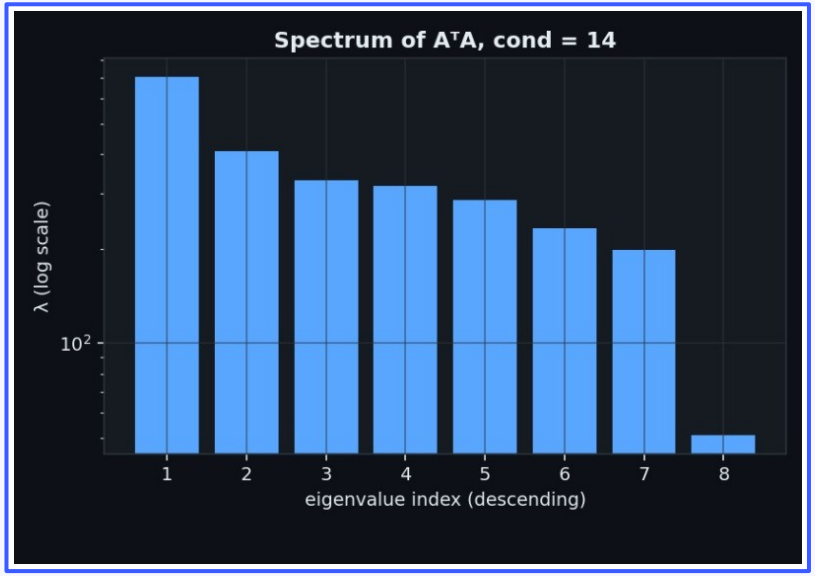
Residual ⊥ feature space.
Choose x that minimizes total squared error.

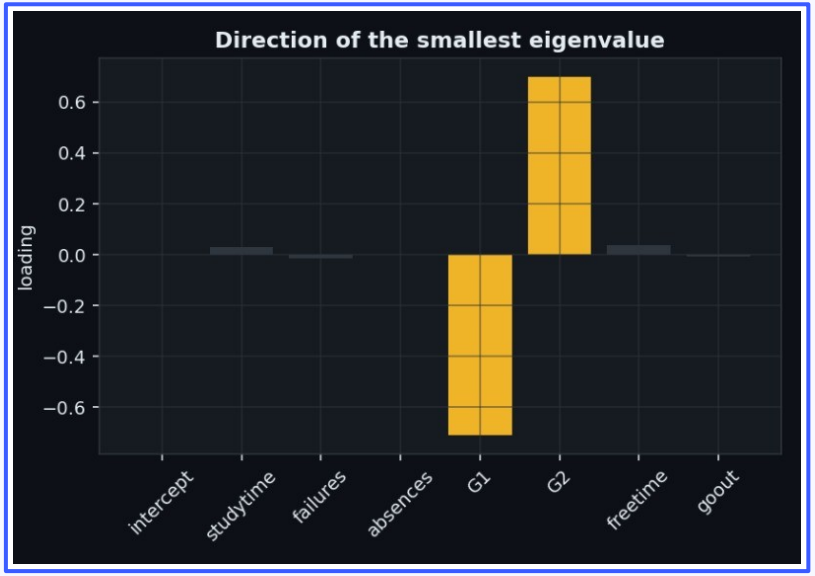


Eigenvalues and Numerical Stability

- With G_sum: one eigenvalue ≈ 0
- After removal: λmin ≈ 50.8, λmax ≈ 706.5
- cond(A^TA) ≈ 13.9 · cond(A) ≈ 3.7

Removing the redundant feature eliminates the near-zero eigenvalue and improves conditioning.

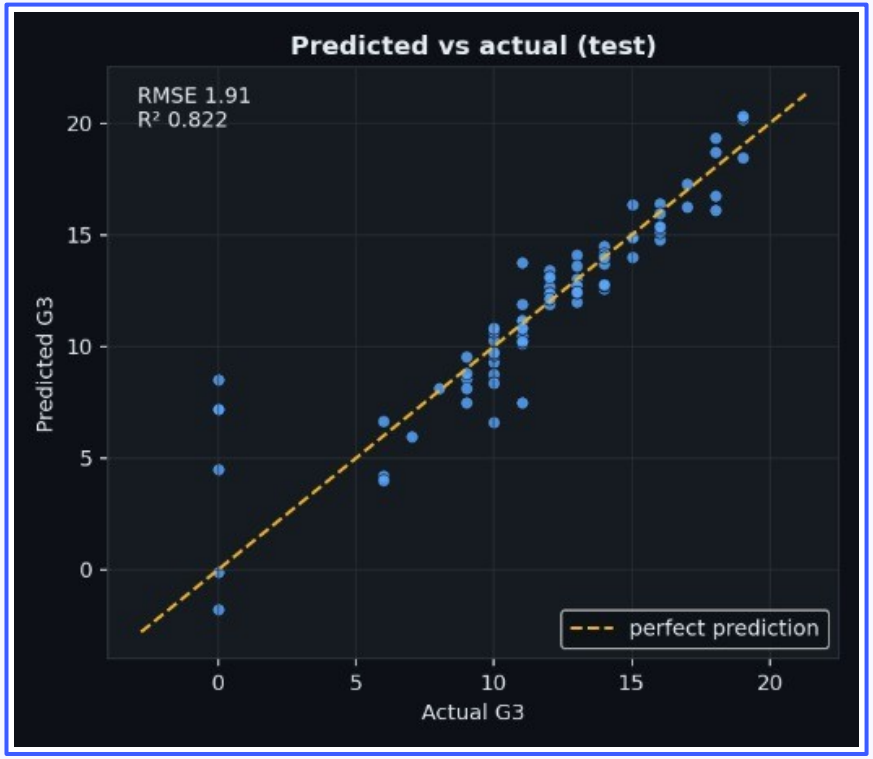




Prediction Results

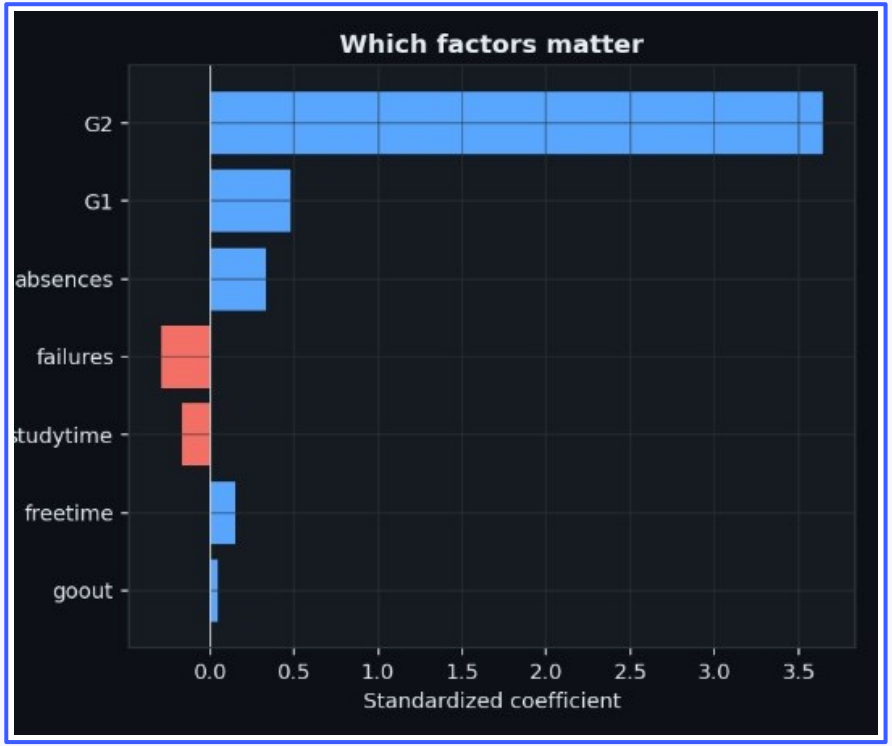
Metric	Training	Testing
RMSE	~1.5–1.6	~1.906
R^2	~0.86–0.88	~0.822

- RMSE: typical error ≈ 1.9 grade points
- R^2: ≈ 82% of variation in final grades explained



Points closer to the diagonal = better predictions

Which Features Influence the Prediction?



- Positive coefficient → raises predicted G3
- Negative coefficient → lowers predicted G3
- Larger |coefficient| → stronger influence

G2 has the strongest positive influence — it captures the student's previous-period performance.

Conclusion

- RREF exposed redundant information
- Rank & nullity: independent directions
- Gram-Schmidt / QR: stable orthonormal basis
- Projection: closest prediction in Col(A)
- Eigenvalues: conditioning & multicollinearity
- Test $R^2 \approx 0.82$ on unseen data

“Linear algebra is not only theoretical — it provides the mathematical foundation for real-world prediction models.”

